# Write your Project title

*By*

Write your Name
CIIT/FA23-RCS-000/ATK
Write your Name
CIIT/FA23-RCS-000/ATK

Project

for

Bachelor of Science in Artificial Intelligence

COMSATS University Islamabad
Attock Campus - Pakistan

SPRING, 2025

**COMSATS University Islamabad, Attock Campus**

# Write your Project Title

This Project is Presented to

COMSATS University Islamabad Attock Campus

In partial fulfillment

of the requirement for the degree of

## BS in Artificial Intelligence

By

Write your Name
CIIT/FA23-RCS-000/ATK
Write your Name
CIIT/FA23-RCS-000/ATK

SPRING, 2025

# Write your Project title

This final year project is submitted to the name of Department of Computer Science as partial fulfillment of the requirement for the award of degree of BS Artificial Intelligence.

| Name | Registration Number |
|---|---|
| Write your Name | CIIT/FA23-RCS-000/ATK |
| Write your Name | CIIT/FA23-RCS-000/ATK |

**Supervisor**

Suervisor Name

Designation,
Department of Computer Science

COMSATS University Islamabad, Attock Campus

Month, Year

# Final Approval

---

This Project titled

## Write your Project Title here

By

*Write your Name Here*
*CIIT/FA23-RCS-000/ATK*
*Write your Name Here*
*CIIT/FA23-RCS-000/ATK*

Has been approved For the COMSATS University Islamabad, Attock Campus.

External Examiner:_____

Dr. External Examinor
University Name

Supervisor:_____

Supervisor Name
Department of Computer Science, (CUI) Attock Campus

HoD:_____

Dr. Zahoor ur Rehman
Department of Computer Science, (CUI) Attock Campus

# Author's Declaration

I "Student's Name", "Student Registration No", hereby declare that I have produced the work presented in this project, during the scheduled period of study. I also declare that I have not taken any material from any source except referred to wherever due to that amount of plagiarism is within an acceptable range. If a violation of HEC rules on research has occurred in this thesis, I shall be liable to punishable action under the plagiarism rules of HEC.

Student Signature

Write your Name

CIIT/FA23-BSAI-000/ATK

Student Signature

Write your Name

CIIT/FA23-BSAI-000/ATK

Date:_____

# Certificate

It is certified that ”Student’s Name”, ”Student Registration No”, has carried out all the work related to this thesis under my supervision at the Department of ”deptt name”, COMSATS University Islamabad, ”campus name” and the work fulfills the requirements for the award of the degree of BS in ”program name”.

Supervisor

_____

Name of Supervisor
Designation,
Department of Computer Science,
CUI, Attock Campus

Head of Department:

_____

Dr. Zahoor ur Rehman
Associate Professor,
Department of Computer Science,
CUI, Attock Campus

*Dedicated to my parents,*

*who have always been my biggest supporters*

*and a constant source of inspiration*

# ACKNOWLEDGEMENTS

**Praise to be ALLAH, the Cherisher and Lord**

**of the World, Most gracious and Most Merciful**

Write Acknowledgements here

Your Name

CIIT/FA23-RCS-000/ATK

# Project Summary

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Maecenas porttitor congue massa. Fusce posuere, magna sed pulvinar ultricies, purus lectus malesuada libero, sit amet commodo magna eros quis urna. Nunc viverra imperdiet enim. Fusce est. Vivamus a tellus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin pharetra nonummy pede. Mauris et orci. Aenean nec lorem. In porttitor. Donec laoreet nonummy augue. Suspendisse dui purus, scelerisque at, vulputate vitae, pretium mattis, nunc. Mauris eget neque at sem venenatis eleifend. Ut nonummy. Fusce aliquet pede non pede. Suspendisse dapibus lorem pellentesque magna. Integer nulla. Donec blandit feugiat ligula. Donec hendrerit, felis et imperdiet euismod, purus ipsum pretium metus, in lacinia nulla nisl eget sapien. Donec ut est in lectus consequat consequat. Etiam eget dui. Aliquam erat volutpat. Sed at lorem in nunc porta tristique. Proin nec augue. Quisque aliquam tempor magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nunc ac magna. Maecenas odio dolor, vulputate vel, auctor ac, accumsan id, felis. Pellentesque cursus sagittis felis. Pellentesque porttitor, velit lacinia egestas auctor, diam eros tempus arcu, nec vulputate augue magna vel risus. Cras non magna vel ante adipiscing rhoncus. Vivamus a mi. Morbi neque.

# TABLE OF CONTENTS

**1 Introduction** . . . . . **1**
1.1 Introduction . . . . . 1
1.2 Significance of the Proposed System . . . . . 1
1.3 Problem Statement . . . . . 2
1.4 Scope and Objectives . . . . . 3
1.5 System Limitations/Constraints . . . . . 3
1.6 Project Deliverables . . . . . 3
1.7 Relevance to Course Modules . . . . . 3
1.8 Contribution . . . . . 4
1.9 Chapter Summary . . . . . 4
**2 Literature Review** . . . . . **5**
2.1 Write first Application problem details or research papers . . . . . 1
2.2 Write second Application problem details or research papers . . . . . 1
2.3 Model-Based Methods . . . . . 1
2.4 Chapter Summary . . . . . 2
**3 Requirement Analysis** . . . . . **3**
3.1 User classes and characteristics . . . . . 1
3.2 Requirement Identifying Technique . . . . . 1
3.3 Functional Requirements . . . . . 1
3.3.1 Functional Requirement X . . . . . 2
3.4 Non-Functional Requirements . . . . . 2
3.4.1 Reliability . . . . . 3
3.4.2 Usability . . . . . 3
3.4.3 Performance . . . . . 3
3.4.4 Security . . . . . 3
3.5 External Interface Requirements . . . . . 3
3.5.1 User Interfaces Requirements . . . . . 4

3.5.2 Software interfaces . . . . . 4
3.5.3 Hardware interfaces . . . . . 5
3.5.4 Communications interfaces . . . . . 5
3.6 Chapter Summary . . . . . 5
**4 Design and Architecture . . . . . 6**
4.1 Modules . . . . . 1
4.1.1 Module 1: Module Name and Description . . . . . 1
4.2 Architectural Design . . . . . 1
4.3 Design Models . . . . . 3
4.3.1 Activity Diagram . . . . . 3
4.3.2 Usecase Diagram . . . . . 3
4.3.3 Class Diagram . . . . . 3
4.3.4 Sequence Diagram . . . . . 3
4.3.5 State Transition Diagram . . . . . 3
4.4 Data Design . . . . . 3
4.4.1 Data Dictionary . . . . . 4
4.5 Human Interface Design . . . . . 4
4.5.1 Screen Images . . . . . 4
4.5.2 Screen Objects and Actions . . . . . 4
4.6 Algorithm . . . . . 4
4.7 Tools and Technologies . . . . . 4
4.8 External APIs/SDKs . . . . . 4
4.9 User Interface . . . . . 5
4.9.1 Login and Home screen . . . . . 5
4.9.2 Assignee Dashboard . . . . . 6
4.10 Deployment . . . . . 6
4.11 Chapter Summary . . . . . 6
**5 Testing Evaluation . . . . . 8**
5.1 Unit Testing . . . . . 1
5.2 Functional Testing . . . . . 1

5.3 Business Rules Testing . . . . . 1
5.4 Integration Testing . . . . . 1
5.5 Chapter Sumamry . . . . . 2
**6 Conclusion and future work** . . . . . **3**
6.1 Conclusion . . . . . 1
6.2 Future Work . . . . . 1
6.3 Chapter Summary . . . . . 2
**References** . . . . . **3**

# LIST OF FIGURES

Fig 4.1 A sample architecture diagram . . . . . 2
Fig 4.2 Caption . . . . . 3
Fig 4.3 A screen shows interface a)login screen and b) menu screen. . . . . 5
Fig 4.4 A screen shows interface of Assignee Dashboard . . . . . 6
Fig 4.5 An example of presenting multiple images in one figure, top is arch 1 and bottom is arch 2 . . . . . 6
Fig 4.6 An example of presenting multiple images in one figure, left is arch 1 and right is arch 2 . . . . . 7
Fig 4.7 An example of presenting multiple images in one figure, use labels A, B, C, D . . . . . 7

# LIST OF TABLES

Table 2.1 Related System Analysis with proposed project solution . . . . . 2
Table 3.1 Description of FR-1 - A multi row table . . . . . 2
Table 4.1 Tools and Technologies for Proposed Project . . . . . 5
Table 4.2 an example of multi column table . . . . . 7
Table 4.3 bullets and numbering in a table . . . . . 7

# Chapter 1

# Introduction

# Chapter Overview

This chapter introduces the background, motivation, and problem statement of the project. It highlights the research gap and explains why this study is important. The chapter also outlines the objectives and structure of the work.

## 1.1 Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Maecenas porttitor congue massa. Fusce posuere, magna sed pulvinar ultricies, purus lectus malesuada libero, sit amet commodo magna eros quis urna. Nunc viverra imperdiet enim. Fusce est. Vivamus a tellus. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin pharetra nonummy pede. Mauris et orci. Aenean nec lorem. In porttitor. Donec laoreet nonummy augue. Suspendisse dui purus, scelerisque at, vulputate vitae, pretium mattis, nunc. Mauris eget neque at sem venenatis eleifend. Ut nonummy. Fusce aliquet pede non pede. Suspendisse dapibus lorem pellentesque magna. Integer nulla. Donec blandit feugiat ligula. Donec hendrerit, felis et imperdiet euismod, purus ipsum pretium metus, in lacinia nulla nisl eget sapien.**??** shows different biometric features.

By comparing the derived model parameters, gait recognition systems can identify individuals based on the unique characteristics of their gait models. Model-based approaches use body parameters as features hence also needing high-resolution videos [**?**]. Overall, model-based gait recognition offers a comprehensive and detailed analysis of gait patterns by incorporating mathematical models. This approach provides valuable insights into the underlying biomechanics of gait and has the potential to improve recognition accuracy in gait recognition systems.

## 1.2 Significance of the Proposed System

Write a concise vision statement that summarizes the long-term purpose, intent and significance of the product. The vision statement should reflect a balanced view that will satisfy the expectations of diverse stakeholders. It can be somewhat idealistic but should be grounded in the realities of existing or anticipated markets. The following keyword tem-

plate works well for crafting a product vision statement:

**For** [target customer]

**Who** [statement of the need or opportunity]

**The** [product name]

**Is** [product category]

**That** [major capabilities, key benefit, compelling reason to buy or use]

**Unlike** [primary competitive alternative, current system, current business process]

**Our product** [statement of primary differentiation and advantages of new product]

Here's a sample vision statement for the Chemical Tracking System, with the keywords in boldface:

***For*** *employees* ***who*** *want to order meals from the company cafeteria or from local restaurants online,* ***the*** *Cafeteria Ordering System is an Internet-based and smartphone-enabled application* ***that*** *will accept individual or group meal orders, process payments, and trigger delivery of the prepared meals to a designated location on the Process Impact campus.* ***Unlike*** *the current telephone and manual ordering processes, employees who use* ***our product*** *(the Cafeteria Ordering System) will not have to go to the cafeteria to get their meals, which will save them time and will increase the food choices available to them.*

This chapter discusses the precise problem to be solved. It should extend to include the outcome.

## 1.3 Problem Statement

Provide a problem statement in a concise paragraph describing

**Why** are you developing this software system?

**What** problem your software system is going to solve?

[Usually in 10-16 sentences]

**Example:**

*Employees at the company Process Impact presently spend an average of 65 minutes per day going to the cafeteria to select, purchase, and eat lunch. About 20 minutes of this time is spent walking to and from the cafeteria, selecting their meals, and paying by cash or credit card. When employees go out for lunch, they spend an average of 90 minutes off-site. Some employees phone the cafeteria in advance to order a meal to be ready for them to pick*

*up. Employees don't always pet the selections they want because the cafeteria runs out of certain items. The cafeteria wastes a significant quantity of food that is not purchased and must be thrown away. These same issues apply to breakfast and supper, although far fewer employees use the cafeteria for those meals than for lunch.*

## 1.4 Scope and Objectives

Write down the scope of your project in a paragraph. Briefly define the main functionalities of the proposed project. Scope defines the boundaries and range of the proposed solution, that what would be the part of your project. Write down in logical flow with consistency. (Usually in 14-18 sentences)

The objectives of our proposed system our three fold.

**Example:**

> [noitemsep,nolistsep,leftmargin=]Reduce the cost of cafeteria food wastage by 40Reduce cafeteria operating costs by 15Increase average effective work time by 15 minutes per cafeteria-using employee per day.

## 1.5 System Limitations/Constraints

Write down the limitations and constraints of the proposed project.

Example:

> [noitemsep,nolistsep,leftmargin=]Some food items that are available from the cafeteria will not be suitable for delivery, so the delivery menus available to patrons of the COS must be a subset of the full cafeteria menus. The COS shall be used only for the cafeteria at the Process Impact campus in Clackamas, Oregon.

## 1.6 Project Deliverables

List down the project deliverables briefly.

- deliverable 1
- deliverable 2

## 1.7 Relevance to Course Modules

A brief explanation of how your project is related to various courses studied during BCS.

## 1.8 Contribution

This study proposes a novel deep learning framework, which utilizes parallel running custom kernels in order to extract dynamic and static feature maps from the input GEI image.

## 1.9 Chapter Summary

The chapter begins by providing an introduction to gait analysis, highlighting its practical applications in real-life scenarios. It then transitions to the various methods utilized for gait recognition. Section 1.1 delves into Appearance-based methods, while section 1.2 focuses on model-based approaches. The subsequent section sheds light on covariate conditions in gait analysis and their impact. Following that, sections 1.4 and 1.5 address the problem statement and research objectives respectively. Lastly, the chapter concludes by discussing the hypothesis proposed as a potential solution to the identified problem.

# Chapter 2

# Literature Review

# Chapter Overview

This chapter introduces the ..... There are different methods based on which Gait analysis has been done in past and can be classified into three types of approaches.

## 2.1 Write first Application problem details or research papers

The problem of known and unknown covariates was tackled by using a simple CNN model for known covariates and employing discriminative and Multi-layer Perceptron as classifiers for unknown covariates [5]. A Gaussian Process classification (GP) architecture was proposed to approximate the viewpoint of each probe gait sequence. At various view angles, gait sequences were captured, and then to measure the similarity between them, Canonical Correlation Analysis (CCA) was used to design the correlation of gait sequences from multiple viewpoints while correlation strength was used as a similarity measure [6].

## 2.2 Write second Application problem details or research papers

Shiraga et al, [12] introduced a network model known as GEInet function. A similar approach was used in [13] which used multiple pooling and convolution layers to extract gait templates [14], which used a framework (TS-CNN) consisting of two-stream CNN architecture which matches mid-level features by learning [15]. Transfer learning uses knowledge from already learned networks, where [**?**] proposed a framework that used transfer [**?**] which was prone to noise in sensory .

## 2.3 Model-Based Methods

A gait recognition approach was introduced in [**?**], also known [**?**] examined three different gait conditions: normal gait, simulated leg length difference, and simulated leg weight difference. Kinematic temporal changes during gait were captured using an online motion recording system. A gait recognition approach was introduced in [**?**], also known [**?**] examined three different gait conditions: normal gait, simulated leg length difference, and simulated leg weight difference. Kinematic temporal changes during gait were captured using an online motion recording system.

Table 2.1 Related System Analysis with proposed project solution

| **Application Name** | **Weakness** | **Proposed Project Solution** |
|---|---|---|
| Row 1, Cell 1 | Row 1, Cell 2 | Row 1, Cell 3 |
| Row 2, Cell 1 | Row 2, Cell 2 | Row 2, Cell 3 |
| Row 3, Cell 1 | Row 3, Cell 2 | Row 3, Cell 3 |
| Row 4, Cell 1 | Row 4, Cell 2 | Row 4, Cell 3 |
| Row 5, Cell 1 | Row 5, Cell 2 | Row 5, Cell 3 |

## 2.4 Chapter Summary

Chapter sumamry comes here

# Chapter 3

# Requirement Analysis

# Chapter Overview

This chapter introduces the background, motivation, and problem statement of the project. It highlights the research gap and explains why this study is important. The chapter also outlines the objectives and structure of the work.

## 3.1 User classes and characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics. *See Appendix A of the word document for an example of how to present user classes and characteristics.*

## 3.2 Requirement Identifying Technique

This section describes the requirements identifying technique(s) which further help to derive functional requirements specification. The selection of the technique(s) will depend on the type of project. For instance,

- Use case (use case diagram + detail use case) is an effective technique for interactive end-user applications.
- Event- response table is for real-time system in which most of the functionalities are performed at backend.
- Storyboarding for graphically intensive applications.

## 3.3 Functional Requirements

This section describes the fr of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depend on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes. For further detail see

Table 3.1 Description of FR-1 - A multi row table

| **Identifier** | **FR-1** |
|---|---|
| **Title** | Title of Requirement |
| **Requirement** | Description of requirement which may be written either from the user or system perspective e.g. If written in a **user perspective:** The [user class or actor name] shall be able to [do something] [to some object] [qualifying conditions, response time, or quality statement]. If written in a **system perspective:** [optional precondition] [optional trigger event] the system shall [expected system response] |
| **Source** | Where this requirement comes from (who originate it) |
| **Rationale** | The motivation behind the requirement |
| **Business Rule (if required)** | Any restriction, policy, the rule that the particular requirement must be fulfilled through its functional behavior |
| **Dependencies** | Requirements ID that is dependent on this requirement |
| **Priority** | High / Medium / Low |

Chapter 10 "Documenting the requirements". Let consider the feature scheme as an example.

### 3.3.1 Functional Requirement X

Itemize the specific functional requirements associated with each feature. These are the software capabilities that must be implemented for the user to carry out the feature's services or to perform a use case. Describe how the product should respond to anticipated error conditions and to invalid inputs and actions. Uniquely label each functional requirement, as described earlier. You can create multiple attributes for each functional requirement, such as rationale, source, dependencies, etc. The following template is required to write functional requirements. For further detail see Chapter 11" Writing excellent requirements".

## 3.4 Non-Functional Requirements

This section specifies nonfunctional requirements other than constraints, which are recorded in section 2.3, and external interface requirements, which will appear in section 7. These quality requirements should be specific, quantitative, and verifiable. Chapter 14 "beyond functionality" presents more information about these quality attribute requirements and

many examples. The following are some examples of documenting guidelines.

### 3.4.1 Reliability

Requirements about how often the software fails. The measurement is often expressed in MTBF (mean time between failures). The definition of a failure must be clear. Also, don't confuse reliability with availability which is quite a different kind of requirement. Be sure to specify the consequences of software failure, how to protect from failure, a strategy for error detection, and a strategy for correction.

### 3.4.2 Usability

Usability requirements deal with ease of learning, ease of use, error avoidance and recovery, the efficiency of interactions, and accessibility. The usability requirements specified here will help the user interface designer create the optimum user experience. Example:

USE-1: The COS shall allow a user to retrieve the previous meal ordered with a single interaction.

### 3.4.3 Performance

State specific performance requirements for various system operations. If different functional requirements or features have different performance requirements, it's appropriate to specify those performance goals right with the corresponding functional requirements, rather than collecting them in this section. Example:

PER-1: 95% of webpages generated by the COS shall download completely within 4 seconds from the time the user requests the page over a 20 Mbps or faster Internet connection.

### 3.4.4 Security

One or more requirements about protection of your system and its data. The measurement can be expressed in a variety of ways (effort, skill level, and time) to break into the system. Do not discuss solutions (e.g. passwords) in a requirements document.

## 3.5 External Interface Requirements

This section provides information to ensure that the system will communicate properly with users and with external hardware or software elements. A complex system with multiple subcomponents should create a separate interface specification or system architecture specification. The interface documentation could incorporate material from other documents by reference. For instance, it could point to a hardware device manual that lists the error codes

that the device could send to the software.

### 3.5.1 User Interfaces Requirements

Describe the logical characteristics of each user interface that the system needs. Some possible items to include are

- References to GUI standards or product family style guides that are to be followed.
- Standards for fonts, icons, button labels, images, color schemes, field tabbing sequences, commonly used controls, and the like.
- Screen layout or resolution constraints.
- Standard buttons, functions, or navigation links that will appear on every screen, such as a help button.
- Shortcut keys.
- Message display conventions.
- Layout standards to facilitate software localization.
- Accommodations for visually impaired users.

Document the user interface design details, such as specific dialog box layouts, in a separate user interface specification, not in the srs . Including screen mock-ups in the srs to communicate another view of the requirements is helpful but make it clear that the mock-ups are not the committed screen designs. If the srs is specifying an enhancement to an existing system, it sometimes makes sense to include screen displays exactly as they are to be implemented. The developers are already constrained by the current reality of the existing system, so it's possible to know up front just what the modified, and perhaps the new, screens should look like.

### 3.5.2 Software interfaces

Describe the connections between this product and other software components (identified by name and version), including other applications, databases, operating systems, tools, libraries, websites, and integrated commercial components (If any). Example:

SI-1: Cafeteria Inventory System SI-1.1: The COS shall transmit the quantities of food items ordered to the Cafeteria Inventory System through a programmatic interface.

### 3.5.3 Hardware interfaces

Describe the characteristics of each interface between the software components and hardware components, if any, of the system. This description might include the supported device types, the data and control interactions between the software and the hardware, and the communication protocols to be used. For more about specifying requirements for systems containing hardware, see Chapter 26, "Embedded and other real-time systems projects."

### 3.5.4 Communications interfaces

State the requirements for any communication functions the product will use, including email, web browser, network protocols, and electronic forms.

Example:

CI-1: The COS shall send an email or text message (based on user account settings) to the Patron to confirm acceptance of an order, price, and delivery instructions.

## 3.6 Chapter Summary

# Chapter 4

# Design and Architecture

# Chapter Overview

This chapter introduces the background, motivation, and problem statement of the project. It highlights the research gap and explains why this study is important. The chapter also outlines the objectives and structure of the work.

## 4.1 Modules

Write down the modules of the proposed project. Each module should highlight features, using bulleted/numbered notation. When developing both a mobile app and a web app, group the modules according to the system types, such as, Client Web App, Client Mobile App, Admin Web App etc.

### 4.1.1 Module 1: Module Name and Description

FE1 Order and pay for meals from the cafeteria menu to be picked up or delivered.

FE2 Order and pay for meals from local restaurants to be delivered.

**You add more module if you required**

## 4.2 Architectural Design

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high-level overview of how the system's modules collaborate with each other in order to achieve the desired functionality. Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together.

Provide a diagram showing the major subsystems and their connections 4.1.

- In initial design stage create Box and Line Diagram for simpler representation of the systems
- After finalizing architecture style/pattern diagram (MVC, Client-Server, Layered, Multi-tiered) create a detailed mapping modules/components to each part of the architecture

*To view example of box and line diagram and architecture styles, see Appendix B. One more thing if your image is not set in relevnt place you can change dimesnion of your images in this code . change the image code parameter instead of image.*

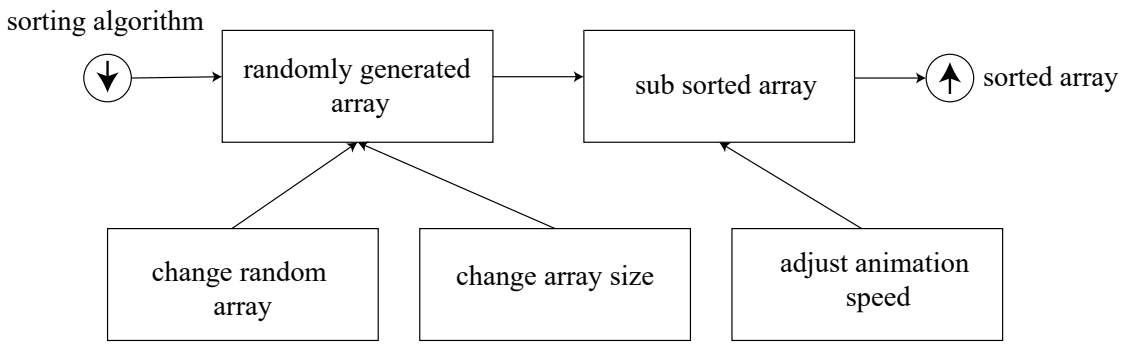



Figure 4.1 A sample architecture diagram

## 4.3 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

### 4.3.1 Activity Diagram

**Activity Diagram**

This section describes the Activity Diagram also add diagrma .

### 4.3.2 Usecase Diagram

Add description and diagram

### 4.3.3 Class Diagram

Add description and diagram

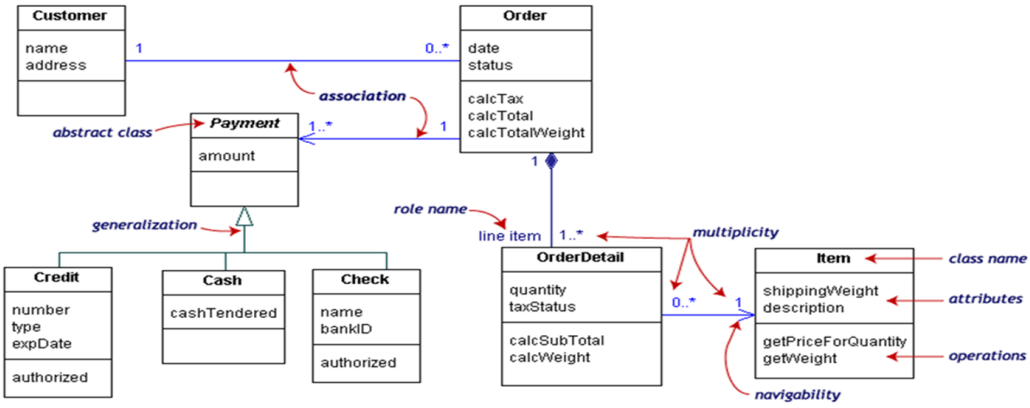



Figure 4.2 Caption

### 4.3.4 Sequence Diagram

Add description and diagram

### 4.3.5 State Transition Diagram

Add description and diagram

## 4.4 Data Design

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed, and organized.

List any databases or data storage items.

### 4.4.1 Data Dictionary

Alphabetically list the system entities or major data along with their types and descriptions. If you provided a functional description, list all the functions and function parameters. If you provided an OO description, list the objects and its attributes, methods and method parameters.

## 4.5 Human Interface Design

Describe the functionality of the system from the user's perspective. Explain how the user will be able to use your system to complete all the expected features and the feedback information that will be displayed for the user.

### 4.5.1 Screen Images

Display screenshots showing the interface from the user's perspective. These can be hand-drawn, or you can use an automated drawing tool. Just make them as accurate as possible. (Graph paper works well.)

### 4.5.2 Screen Objects and Actions

A discussion of screen objects and actions associated with those objects.

## 4.6 Algorithm

Mention the algorithm(s) used in your project to get the work done with regards to major modules. Provide a pseudocode explanation regarding the functioning of the core features. Be sure to use the correct syntax and semantics for algorithm representations. Following are few examples of algorithms/pseudocode:

## 4.7 Tools and Technologies

Mention all the hardware/software tools and technologies with version number which will be used in implementation of the project. Write about the APIs, language(s), SDK(s) etc. which you will use for implementation.

## 4.8 External APIs/SDKs

Describe the third-party APIs/SDKs used in the project implementation in the following table. Few examples of APIs are provided in the table.

Table 4.1 Tools and Technologies for Proposed Project

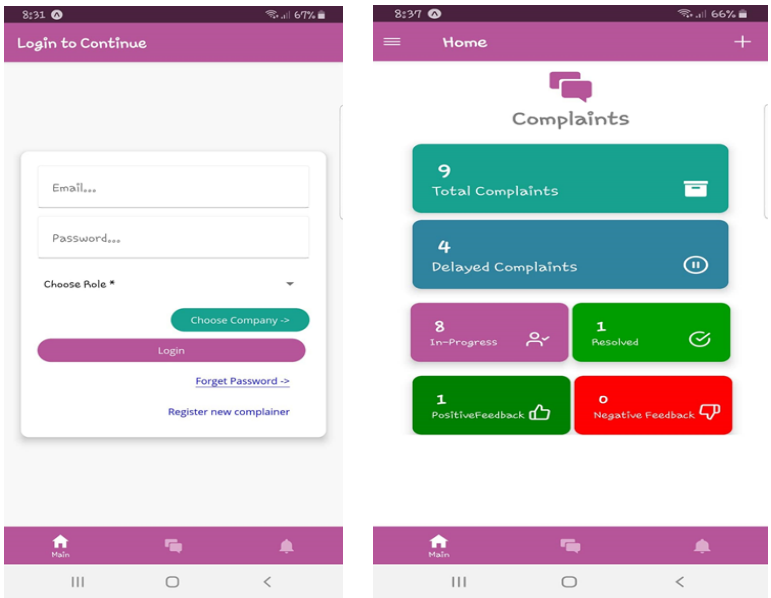
| Tools and Technologies | Tools | Version | Rational |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

## 4.9 User Interface

Details about user interface with descriptions. Provide the User Interface for each sub-system (such as Mobile App, Web App, Client App, Admin App). Provide description of each User Interface explaining the details. When inserting User Interfaces, use appropriate size of the image, for example, for mobile app, 2-4 screens can be placed on a single page. Following are few examples of User Interfaces:

### 4.9.1 Login and Home screen

Login screen of our mobile app where user have to choose its role and its company. Home screen where total, delayed and other complaints are shown. 4.3




(a) (b)

Figure 4.3 A screen shows interface a)login screen and b) menu screen.

### 4.9.2 Assignee Dashboard

Complain Assignee can view the graphs of month-wise complains, Resolved complains, summary and a list of submitted complains 4.4.

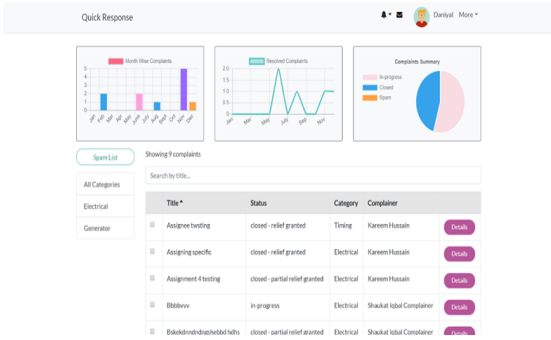



Figure 4.4 A screen shows interface of Assignee Dashboard

## 4.10 Deployment

Specify the deployment environments used for hosting and live testing of all the sub-systems of the project. Provide the details of hosting/cloud service used, set of available software and their versions used etc.

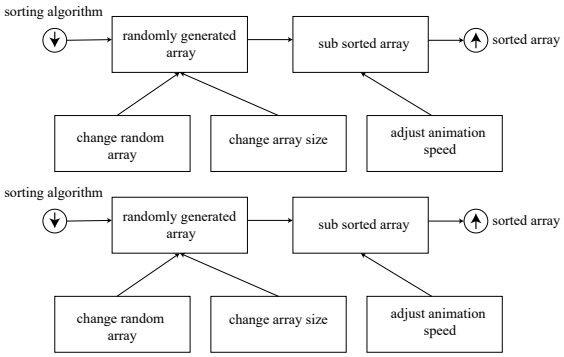



Figure 4.5 An example of presenting multiple images in one figure, top is arch 1 and bottom is arch 2

## 4.11 Chapter Summary

Chapter summary comes here

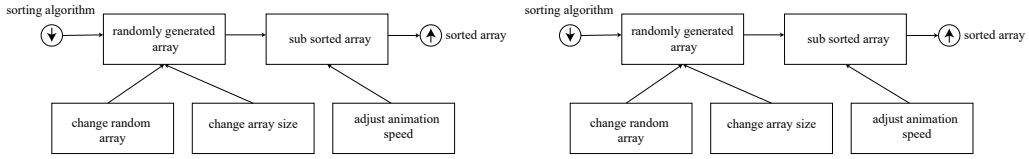



Figure 4.6 An example of presenting multiple images in one figure, left is arch 1 and right is arch 2

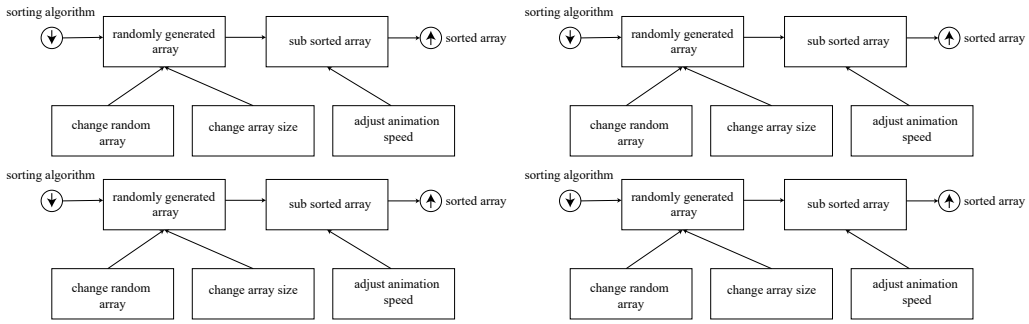



Figure 4.7 An example of presenting multiple images in one figure, use labels A, B, C, D

Table 4.2 an example of multi column table

| combine columns | | combine columns | |
|---|---|---|---|
| x | y | x | y |
| 1<br>1<br>1 | 2<br>2<br>2 | 3<br>3<br>3 | 4<br>4<br>4 |

Table 4.3 bullets and numbering in a table

| columns 1 | columns 2 | columns 3 | columns 4 |
|---|---|---|---|
| [leftmargin=10px]point 1 point 2 | [leftmargin=10px]point 1 point 2 | x | y |
| 1<br>1<br>1 | 2<br>2<br>2 | 3<br>3<br>3 | 4<br>4<br>4 |

# Chapter 5

# Testing  Evaluation

# Chapter Overview

This chapter introduces the background, motivation, and problem statement of the project. It highlights the research gap and explains why this study is important. The chapter also outlines the objectives and structure of the work.

## 5.1 Unit Testing

It's a level of software testing where individual units of a software/component are tested. The purpose is to validate that each unit of the software performs as designed.

Unit Testing 1: Login as Patient with valid and invalid credentials

Testing Objective: To ensure the login form is working correctly with valid and invalid credentials/inputs.

## 5.2 Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the module is tested. This is to ensure that the system produced meets the specifications and requirements.

Functional Testing 1: Login with different roles (Management, Patient, Doctor)

Objective: To ensure that the correct page with the correct navigation bar is loaded.

## 5.3 Business Rules Testing

Decision table based testing technique is used to test business rules. The business rules were defined in FRs and Use Cases Decision based testing uses a systematic approach where input and outputs are provided in tabular form. It is a precise and compact way to model complicated logic. The table contains conditions and actions are used for test cases where conditions as inputs and actions as outputs.

Detailed example is as given in Appendix E. see reference [**?**]

## 5.4 Integration Testing

Integration tests assess whether a set of classes that must work together do so without error. They ensure that the interfaces and linkages between different parts of the system work properly. At this point, the classes have passed their individual unit tests, so the focus

now is on the flow of control among the classes and on the data exchanged among them. Integration testing follows the same general procedures as unit testing: The tester develops a test plan that has a series of tests, which, in turn, have a test. Integration testing is often done by a set of programmers and/or systems analysts.

Integration Testing 1: Scheduling Patient Appointment

Testing Objective: To ensure the scheduling is being done correctly and the interface between module 'Patient/Doctor Management' and module 'Appointment/Scheduling' is running correctly.

## 5.5 Chapter Sumamry

chapter summary goes here

# Chapter 6

# Conclusion and future work

# Chapter Overview

This chapter introduces the background, motivation, and problem statement of the project. It highlights the research gap and explains why this study is important. The chapter also outlines the objectives and structure of the work.

## 6.1 Conclusion

This chapter summarizes the major contributions and findings of this project. The primary objective of this work was to design and develop an automated spot-specific spraying system for weed detection in wheat crops. The proposed approach successfully achieved this objective by combining machine learning techniques with image processing to detect and localize weeds in real time.

The results demonstrated that the proposed system was able to accurately identify weed-infested regions and perform precise spot-specific spraying. The outcomes confirmed that the methodology reduced unnecessary pesticide usage while maintaining effectiveness. The findings highlight the significance of this work in promoting sustainable agriculture by reducing chemical consumption, lowering costs, and minimizing environmental impact.

## 6.2 Future Work

Although the project achieved its intended objectives, there are several directions for future research and improvement:

- **Dataset Expansion:** Incorporate a larger and more diverse dataset covering different crop growth stages, weed types, and field conditions.
- **Model Enhancement:** Explore advanced deep learning architectures, such as transformers or reinforcement learning, to improve detection accuracy and decision-making.
- **Hardware Optimization:** Implement more efficient spraying mechanisms and integrate lightweight hardware for real-time field deployment.
- **Scalability:** Extend the system for other crops beyond wheat, and test it in large-scale farming scenarios.

- **Integration with Smart Agriculture:** Combine the spraying system with IoT sensors, drones, or autonomous ground robots for fully automated precision farming solutions.

In conclusion, while the proposed system provides a strong foundation for automated weed management, future enhancements will enable more robust, scalable, and intelligent solutions for precision agriculture.

## 6.3 Chapter Summary

# References

[1] C. Shen, S. Yu, J. Wang, G. Q. Huang, and L. Wang, "A Comprehensive Survey on Deep Gait Recognition: Algorithms, Datasets and Challenges," 2022, doi: 10.48550/ARXIV.2206.13732.

[2] J. Han and B. Bhanu, "Individual recognition using gait energy image," IEEE Trans. Pattern Anal. Mach. Intell., vol. 28, no. 2, pp. 316–322, Feb. 2006, doi: 10.1109/TPAMI.2006.38.

[3] H. Iwama, M. Okumura, Y. Makihara, and Y. Yagi, "The OU-ISIR Gait Database Comprising the Large Population Dataset and Performance Evaluation of Gait Recognition," IEEE Trans. Inf. Forensics Secur., vol. 7, no. 5, pp. 1511–1521, Oct. 2012, doi: 10.1109/TIFS.2012.2204253.

[4] Shuai Zheng, Junge Zhang, Kaiqi Huang, Ran He and Tieniu Tan. Robust View Transformation Model for Gait Recognition. Proceedings of the IEEE International Conference on Image Processing, 2011

[5] M. Bukhari et al., "An Efficient Gait Recognition Method for Known and Unknown Covariate Conditions," IEEE Access, vol. 9, pp. 6465–6477, 2021, doi: 10.1109/ACCESS.2020.3047266.

[6] K. Bashir, T. Xiang, and S. Gong, "Cross View Gait Recognition Using Correlation Strength," in Procedings of the British Machine Vision Conference 2010, Aberystwyth, 2010, p. 109.1-109.11. doi: 10.5244/C.24.109.

[7] R. Martín-Félez and T. Xiang, "Uncooperative gait recognition by learning to rank," Pattern Recognit., vol. 47, no. 12, pp. 3793–3806, Dec. 2014, doi: 10.1016/j.patcog.2014.06.010.

[8] Xuelong Li, S. Lin, Shuicheng Yan, and Dong Xu, "Discriminant Locally Linear Embedding With High-Order Tensor Data," IEEE Trans. Syst. Man Cybern. Part B Cybern., vol. 38, no. 2, pp. 342–352, Apr. 2008, doi: 10.1109/TSMCB.2007.911536.

[9] Dong Xu, Shuicheng Yan, Dacheng Tao, Lei Zhang, Xuelong Li, and Hong-Jiang Zhang, "Human Gait Recognition With Matrix Representation," IEEE Trans. Circuits Syst. Video Technol., vol. 16, no. 7, pp. 896–903, Jul. 2006, doi: 10.1109/TCSVT.2006.877418.

[10] D. Tao, X. Li, X. Wu, and S. J. Maybank, "General Tensor Discriminant Analysis and Gabor Features for Gait Recognition," IEEE Trans. Pattern Anal. Mach. Intell., vol. 29, no. 10, pp. 1700–1715, Oct. 2007, doi: 10.1109/TPAMI.2007.1096.

[11] H. Lee, S. Hong, and E. Kim, “An Efficient Gait Recognition with Backpack Removal,” EURASIP J. Adv. Signal Process., vol. 2009, no. 1, p. 384384, Dec. 2009, doi: 10.1155/2009/384384.

[12] K. Shiraga, Y. Makihara, D. Muramatsu, T. Echigo, and Y. Yagi, “GEINet: View-invariant gait recognition using a convolutional neural network,” in 2016 International Conference on Biometrics (ICB), Halmstad, Sweden, Jun. 2016, pp. 1–8. doi: 10.1109/ICB.2016.7550060.

[13] H. Chao, K. Wang, Y. He, J. Zhang, and J. Feng, “GaitSet: Cross-view Gait Recognition through Utilizing Gait as a Deep Set,” IEEE Trans. Pattern Anal. Mach. Intell., pp. 1–1, 2021, doi: 10.1109/TPAMI.2021.3057879.

[14] Z. Wu, Y. Huang, L. Wang, X. Wang, and T. Tan, “A Comprehensive Study on Cross-View Gait Based Human Identification with Deep CNNs,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 39, no. 2, pp. 209–226, Feb. 2017, doi: 10.1109/TPAMI.2016.2545669.

[15] C. Fan et al., “GaitPart: Temporal Part-Based Model for Gait Recognition,” in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, Jun. 2020, pp. 14213–14221. doi: 10.1109/CVPR42600hing invariant human gait recognition using modified local optimal oriented pattern binary descriptor. Multimed Tools Appl 79, 2873–2896 (2020), doi: 10.1007/s11042-019-08400-8